

Fault Lines of Inequality: When Earthquakes Discriminate

COM-480 Data Visualization - Milestone 2

Website: com-480-data-visualization.github.io/the-visualizers/

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1 Project Goal

In January 2010 a M7.0 earthquake killed more than 222,000 people in Haiti. Seven weeks later a M8.8 earthquake, releasing roughly $500\times$ more energy, struck Chile and killed fewer than 600. Our project is a **scrollytelling interactive website** that uses this contrast to argue that earthquake mortality is driven by wealth, infrastructure, and preparedness rather than seismic force alone. We combine the NOAA Significant Earthquake Database (2,679 events, 1960-2026) with World Bank and UNDP indicators (GDP per capita, HDI, hospital beds, urbanisation, population density) to take the reader from a global overview down to country-level comparison.

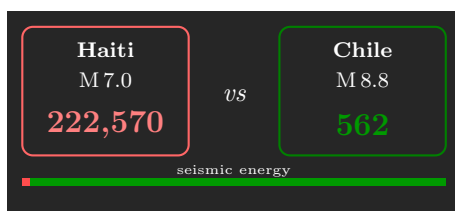
The site is organised as a single scrolling page with a sticky top navigation, a scroll-progress bar, and seven sections: a hook, five scroll-driven visualisations, and an exploratory country-comparison tool.

2 Visualization Design and Sketches

Each section below is described with a sketch, the intent, the tools and course lectures it draws on.

2.1 V1 - Core Visualization & The Hook (Haiti vs. Chile)

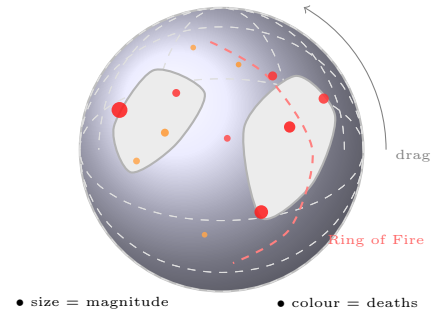
Two cards on a dark background show flag, magnitude, and a counter indicating the deaths. Between them, a horizontal bar encodes the seismic-energy ratio (Chile's fill is $500\times$ wider). A short caption below asks how the smaller quake caused $396\times$ more deaths, introducing the reader to the paradox that the rest of the site will explain.



Tools: HTML/CSS, D3 tween counters, IntersectionObserver. **Lectures:** *Designing Viz*; *JavaScript*; *D3*.

2.2 V2 - Interactive Globe

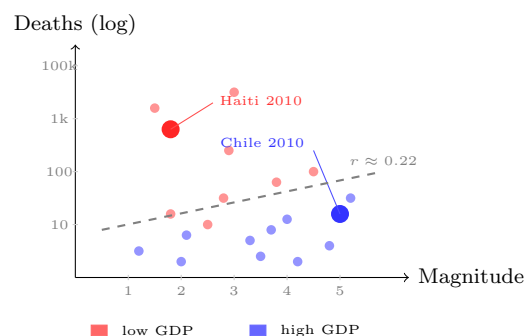
An orthographic globe showing every earthquake as a red dot. Four scroll steps: (1) uniform dots, (2) radius $\propto$ magnitude, (3) recolor by death counter, (4) auto-rotate to frame the Pacific Ring of Fire. Dragging rotates the globe; hovering shows a tooltip with info about that earthquake.



Tools: D3 geoOrthographic, TopoJSON, D3 drag, d3.timer. **Lectures:** *Maps*; *Interactions*; *Marks & Channels*.

2.3 V3 - The Magnitude Myth (Scatter)

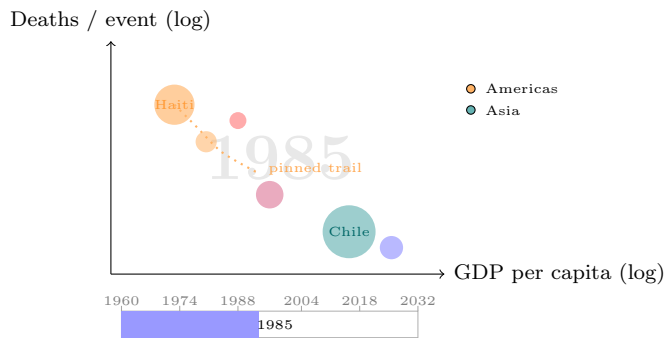
Magnitude on x , deaths on y (log). Scroll steps progressively (1) reveal the dots, (2) overlay a regression line showing the weak fit, (3) recolor each dot by GDP per capita on a red-blue diverging scale, (4) annotate Haiti and Chile 2010. The chart shows that wealth explains deaths better than magnitude does.



Tools: D3 log/diverging scales, JS linear regression, shared annotation and tooltip components. **Lectures:** *Tabular Data*; *Marks & Channels*; *D3*; *Storytelling*.

2.4 V4 - Gapminder Bubble Chart

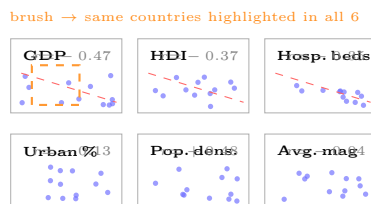
GDP per capita (x , log) vs. deaths per event (y , log), animated across decades. Bubble area encodes event count, colour encodes world region. The reader uses the play button (or drags the year slider) to move through time and watch the bubbles shift. Clicking a bubble keeps it selected and draws a line showing where that country was in previous years.



Tools: D3 log/sqrt scales, `d3.interval`, `d3.line` with `curveCatmullRom` for trails. **Lectures:** *Tabular Data*; *Interactions*; *More Interactive D3*.

2.5 V5 - Correlation Small Multiples

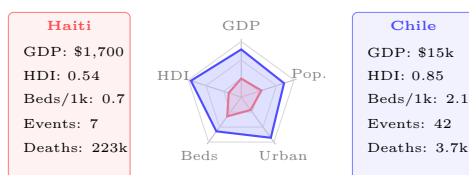
A 3×2 grid of scatter plots, each pairing one socio-economic indicator (GDP, HDI, hospital beds, urbanisation, population density, average magnitude) with deaths per event and labelled with its Pearson r . A brush on any panel cross-highlights the same countries in all six, letting the reader spot who is consistently vulnerable across axes.



Tools: D3 linked brushes, Pearson r in JS. **Lectures:** *Tabular Data* (small multiples); *Interactions* (brush & link); *More Interactive D3*.

2.6 V6 - Country Comparison Tool

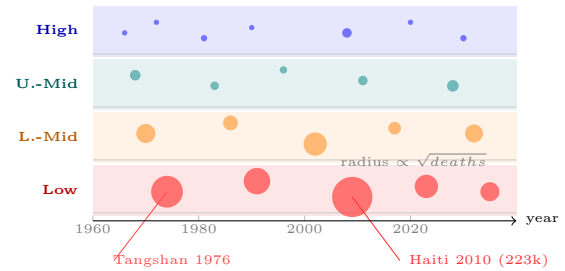
The only fully free-exploration section. Two dropdowns (plus one-click suggestion chips for canonical pairs like Haiti-Chile and Iran-Japan) drive three linked views: stat cards, a radar chart overlaying normalised indicators for both countries, and a per-country event timeline.



Tools: D3 radial paths, HTML `<select>` with live filtering. **Lectures:** *Interactions* (details on demand); *Designing Viz*; *Marks & Channels*.

2.7 V7 - Income-Group Timeline

Four horizontal bands (High / Upper-Middle / Lower-Middle / Low income). Each event is a circle at its year with vertical jitter, sized by deaths and coloured by income group. Scroll steps annotate the deadliest outliers and make the visual claim of the piece: the biggest circles all sit in the bottom bands.



Tools: D3 `scaleBand`, `scaleSqrt`, reusable annotation component. **Lectures:** *Tabular Data*; *Storytelling*; *Marks & Channels*.

3 Tools Summary

D3.js v7 powers every chart (scales, axes, transitions, geo projections, brushes, drag, animations). **TopoJSON** supplies world boundaries for the globe. **Scrollama** drives scroll-triggered steps in V2-V5 and V7. **Vite** provides the dev server and production bundling. **HTML/CSS** handles layout, sticky nav, and progress bar. **Python/pandas** pre-merges the NOAA, World Bank, and UNDP tables into the JSON files consumed by the site.

4 Implementation Breakdown

4.1 Core Visualisation (Minimum Viable Product)

These components carry the argument on their own:

1. **Hook (V1)** - frames Haiti vs. Chile.
2. **Magnitude Myth scatter (V3)** - shows wealth explains deaths better than magnitude.
3. **Correlation small multiples (V5)** - multi-indicator evidence with brush-and-link.
4. **Income-group timeline (V7)** - visual summary of the disparity.
5. The Scrollama-driven narrative that connects them.

4.2 Enhancements (Can Be Dropped Without Losing Core Message)

1. **Interactive globe (V2)** - nice to look at, but the argument works without it.
2. **Gapminder animation (V4)** - compelling temporal view, partly redundant with V3 and V5.
3. **Country comparison tool (V6)** - free exploration outside the linear narrative.
4. **Trajectory trails (V4)** and **radar overlay (V6)** - polish features for the two enhancements above.

4.3 Data Pipeline

Raw inputs are the NOAA Significant Earthquake Database (CSV, one row per event) and the World Bank + UNDP indicator tables (wide-format CSVs, one row per country-year). A pandas script in the repository joins events to country-year indicators on (`country`, `year`), derives per-event fields (deaths, magnitude, GDP at event time, income group), and writes five compact JSON artefacts that the site consumes: `earthquakes.json` (events), `countries.json` (country metadata), `country_year.json` (time-series indicators for V4), `paired_events.json` (country-pair data for V6), and `world-110m.json` (TopoJSON for V2).

4.4 Narrative Arc

The sections are ordered to move the reader from *anecdote* (V1: Haiti vs Chile) to *evidence* (V3: scatter, V5:

small multiples) to *mechanism* (V4: time-animated Gap-minder) and finally *summary* (V7: income-group timeline). V2 (globe) and V6 (comparison tool) are optional:

V2 adds geographic context, V6 lets the reader explore on their own.